

CG3CG3 — Chapters 3 & 4

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2026-05-13

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CHAPTER 3: METHODOLOGY

3.1 Research Design

This study adopts a quantitative, non-experimental research design using secondary data obtained from the Kaggle Developer Survey dataset. The research is descriptive and predictive in nature, aiming to identify factors that significantly influence developer compensation and to construct a reliable predictive model for annual salary.

The analytical approach is grounded in multiple linear regression, a classical statistical method suited for estimating the linear relationship between a continuous dependent variable and multiple independent predictors. Because the data are observational rather than experimentally controlled, the study focuses on identifying statistically significant associations rather than making causal claims.

Model validity is assessed through a comprehensive suite of diagnostic tests covering the core OLS assumptions: normality of residuals, homoscedasticity, independence of errors, and absence of severe multicollinearity. Multiple model specifications are compared using Adjusted R^2 , AIC, and BIC to arrive at the most parsimonious and well-fitting final model.

3.2 Data Collection

The dataset used in this study was sourced from Kaggle, a publicly accessible platform for open datasets widely used in academic and data science research. The dataset is a **synthetically generated** developer survey containing 12,000 records and 32 variables designed to simulate realistic developer demographics, compensation, and technology adoption patterns.

Dataset Source: <https://www.kaggle.com/datasets/meruvakodandasuraj/developer-survey-2026-skills-salary-and-ai-impact>

The dataset was loaded into R using the following command:

Table 1: Table 1. Developer Demographics

respondent_id	survey_year	country	age	gender	education_level	years_of_experience	role
R00001	2024	India	43	Man	Some college	22	Front-End Developer
R00002	2025	Italy	28	Man	Master's degree	7	Machine Learning Engineer
R00003	2021	South Africa	23	Woman	Master's degree	0	Machine Learning Engineer
R00004	2023	India	22	Prefer not to say	Bachelor's degree	1	QA / Test Engineer
R00005	2025	United States	34	Man	Bachelor's degree	11	Front-End Developer
R00006	2021	India	33	Man	Bachelor's degree	15	Front-End Developer
R00007	2022	China	29	Man	Bachelor's degree	13	Cloud Engineer
R00008	2022	India	31	Man	Bachelor's degree	12	Full-Stack Developer
R00009	2025	United States	46	Man	Bachelor's degree	26	Front-End Developer
R00010	2020	Indonesia	37	Man	Self-taught / Bootcamp	11	Front-End Developer
R00011	2021	Pakistan	20	Prefer not to say	Master's degree	0	Embedded Developer
R00012	2026	United States	36	Woman	Master's degree	16	QA / Test Engineer
R00013	2022	South Africa	50	Man	Self-taught / Bootcamp	25	Full-Stack Developer
R00014	2025	Belgium	24	Man	Bachelor's degree	0	Front-End Developer
R00015	2021	India	21	Man	Master's degree	2	Student / Intern

Table 2: Table 2. Employment and Salary Information

employment_status	company_size	work_mode	coding_hours_per_week	annual_salary_usd	job_satisfaction	career_satisfaction
Unemployed	1-10	Hybrid	12	3700	1	4
Unemployed	201-1000	Remote	26	9660	3	4
Full-time	1001-5000	In-Office	43	26200	4	2
Unemployed	10000+	Hybrid	27	0	5	2
Full-time	51-200	Remote	25	172100	3	4
Full-time	51-200	In-Office	29	14900	4	4
Full-time	1001-5000	Hybrid	27	45200	1	5
Full-time	51-200	Remote	34	16200	5	4
Full-time	201-1000	Hybrid	40	245100	4	5
Full-time	1001-5000	In-Office	45	17900	2	4
Student	5001-10000	Hybrid	12	0	4	3
Full-time	11-50	In-Office	33	211500	3	4
Full-time	1-10	Hybrid	35	46500	5	4
Full-time	5001-10000	Remote	41	34600	4	4
Full-time	201-1000	Hybrid	51	4600	4	5

An initial inspection confirmed that only one column, `ai_tools_used`, contained missing values (approximately 19.8% of rows). All other columns were complete prior to imputation. The `ai_tools_used` column was excluded from the model as it is a semicolon-delimited multi-value free-text field not directly suitable for regression encoding.

Table 3: Table 3. Technology Stack

primary_os	languages_used	frameworks_used	databases_used	cloud_platforms_used	ide_editor
macOS	Swift;JavaScript;Python;HTML/CSS;SQL;Kotlin	Node.js;Ruby on Rails	PostgreSQL;Firebase Realtime DB;SQLite;Neo4j	AWS;Cloudflare;Hetzner	Zed
macOS	Python;HTML/CSS;Java	FastAPI	PostgreSQL;SQLite;MongoDB;MySQL	Cloudflare	Zed;Visual Studio;IntelliJ IDEA
Linux	Swift;Python;Java;JavaScript;Bash/Shell;SQL	Spring Boot	DynamoDB	Google Cloud;Railway	VS Code;Visual Studio
Linux	Python;JavaScript;TypeScript	Express	PostgreSQL;Oracle;Supabase	DigitalOcean;Netlify;Vercel	Vim/Neovim;VS Code
Windows	C++;JavaScript;Scala;TypeScript	Django;.NET;Vue.js;Next.js;React	Firebase Realtime DB;SQLite;Redis;ClickHouse	Cloudflare;Vercel	VS Code;Vim/Neovim;Visual Studio
Windows	JavaScript;C;Python;SQL	React;Node.js;Laravel;Express;Flutter	SQLite;Microsoft SQL Server;PostgreSQL	Heroku	WebStorm;Visual Studio
Linux	PHP;Bash/Shell;SQL;Python	Spring Boot;htmx;Tailwind CSS;Bootstrap	MySQL	AWS;Azure	VS Code;Cursor
Windows	C;Kotlin;Bash/Shell;SQL;JavaScript	FastAPI;React	Microsoft SQL Server	AWS;Azure	VS Code;Vim/Neovim
macOS	JavaScript	Flask;Next.js;TensorFlow	Redis;PostgreSQL;Supabase;MySQL	AWS	Visual Studio;IntelliJ IDEA;VS Code
Windows	C#;Rust;HTML/CSS	Vue.js;Bootstrap	DynamoDB;PostgreSQL;CockroachDB;MariaDB	DigitalOcean;Vercel	Android Studio;Xcode;VS Code
macOS	SQL;JavaScript	Django;Flask;Angular;Next.js	PostgreSQL;Microsoft SQL Server	Azure;DigitalOcean	WebStorm;PyCharm;Vim/Neovim
macOS	SQL;Go;Python;JavaScript	TensorFlow;Tailwind CSS;NestJS	Firebase Realtime DB;PostgreSQL	Fly.io	Sublime Text;IntelliJ IDEA
Windows	PHP;HTML/CSS;TypeScript;SQL	Spring Boot;Next.js	Cassandra;Neo4j;MySQL;PostgreSQL	AWS	VS Code;IntelliJ IDEA;Cursor
Windows	Zig;Python;JavaScript;C++	Angular	SQLite;PostgreSQL	Google Cloud;Azure;Netlify	IntelliJ IDEA;Android Studio
Windows	PHP	Node.js;jQuery;FastAPI;Spring Boot	Microsoft SQL Server;Redis;SQLite;DynamoDB	Cloudflare	VS Code;Cursor;Vim/Neovim

Table 4: Table 4. AI Usage and Perception

ai_tools_used	ai_usage_frequency	ai_productivity_boost_pct	ai_sentiment	ai_job_threat_perception
ChatGPT	Weekly	14	Somewhat positive	Minimal threat
	Weekly	8	Somewhat positive	Minimal threat
	Rarely	1	Very negative	Significant threat
	Multiple times daily	61	Neutral	Minimal threat
	Weekly	15	Neutral	Moderate threat
	Daily	36	Very positive	Significant threat
ChatGPT	Monthly	8	Somewhat positive	Minimal threat
	Multiple times daily	16	Neutral	No threat
	Weekly	26	Somewhat positive	Will replace most devs
	Weekly	19	Very negative	Significant threat
ChatGPT;Cursor AI	Weekly	19	Very negative	Significant threat
Cursor AI;ChatGPT;Claude	Never	9	Somewhat positive	Moderate threat
ChatGPT	Multiple times daily	36	Very positive	No threat
Claude Code;Perplexity	Monthly	4	Very positive	Minimal threat
	Daily	7	Very positive	Moderate threat
	Multiple times daily	45	Very positive	Minimal threat

Table 5: Table 5. Career Growth and Preferences

primary_learning_source	open_source_contributor	looking_for_job	remote_work_preference_1to5	burnout_frequency	want_to_learn_next
Online courses	1	Actively looking	2	Sometimes	Elixir;Zig;AI/ML
Stack Overflow	1	Not looking	4	Often	Kubernetes;Elixir;Go
YouTube	0	Open to offers	5	Often	Go
Stack Overflow	1	Open to offers	3	Rarely	HTMX;Swift;Rust
Official docs	0	Open to offers	1	Sometimes	TypeScript
Stack Overflow	0	Not looking	3	Sometimes	Kubernetes;TypeScript
Official docs	0	Not looking	5	Often	AI/ML;Go
Books	1	Not looking	5	Often	Kotlin;Swift;Kubernetes
YouTube	0	Not looking	5	Often	Swift;HTMX
ChatGPT/AI	1	Actively looking	4	Rarely	Rust;Kotlin;Go
Colleagues	0	Not looking	2	Always	Rust
Stack Overflow	0	Not looking	5	Rarely	Elixir;Zig;HTMX
Blogs	1	Open to offers	3	Always	Go;Elixir
ChatGPT/AI	1	Open to offers	4	Sometimes	TypeScript;Swift
Official docs	0	Open to offers	4	Often	Python;HTMX

3.3 Variables Description

The study models annual developer salary as the dependent variable, with a comprehensive set of demographic, employment, educational, and behavioral predictors serving as indepen-

dent variables.

3.3.1 Dependent Variable

The dependent variable is `log_salary`, defined as the natural logarithm of `annual_salary_usd`. A log transformation was applied because raw salary distributions are strongly right-skewed; the transformation stabilizes variance across fitted values and brings the distribution closer to normality, both of which are assumptions of OLS regression. Respondents with `annual_salary_usd` ≤ 0 were excluded from the analysis as these entries reflect unemployed or unreported salary statuses rather than true compensation values.

3.3.2 Independent Variables

The independent variables include all numeric, ordinal, and nominal features available in the cleaned dataset after excluding ID columns, multi-value text fields, and the target variable. Table 3.1 presents the full set of predictors, their measurement type, and how each was encoded for regression.

Variable	Type	Encoding / Notes
<code>years_of_experience</code>	Continuous numeric	Raw numeric; direct linear predictor of salary
<code>age</code>	Continuous numeric	Raw numeric
<code>coding_hours_per_week</code>	Continuous numeric	Raw numeric
<code>ai_productivity_boost_pct</code>	Continuous numeric	Raw numeric; self-reported % boost from AI tools
<code>remote_work_preference_1to5</code>	Ordinal numeric	Raw numeric (1 = fully on-site, 5 = fully remote)
<code>open_source_contributor</code>	Binary numeric	Raw numeric (0/1)
<code>education_level</code>	Ordered factor	7-level ordered factor from High school to Doctorate (PhD)
<code>company_size</code>	Ordered factor	7-level ordered factor from 1–10 to 10000+ employees
<code>burnout_frequency</code>	Ordered factor	5-level ordered factor: Never to Always
<code>ai_usage_frequency</code>	Ordered factor	6-level ordered factor: Never to Multiple times daily
<code>role</code>	Nominal factor	Unordered; each level gets a dummy coefficient vs. baseline

Variable	Type	Encoding / Notes
employment_status	Nominal factor	Unordered; e.g., Full-time, Part-time, Freelance
work_mode	Nominal factor	Unordered; e.g., Remote, Hybrid, In-Office
gender	Nominal factor	Unordered
primary_os	Nominal factor	Unordered; e.g., Windows, macOS, Linux
ai_sentiment	Nominal factor	Unordered; attitude toward AI in the workplace
ai_job_threat_perception	Nominal factor	Unordered; perceived job threat level from AI
primary_learning_source	Nominal factor	Unordered; e.g., Official docs, Stack Overflow, YouTube
looking_for_job	Nominal factor	Unordered; current job-seeking status
country	Nominal factor	Unordered; country of employment

Table 3.1. Independent Variables Used in the Multiple Linear Regression Model

3.4 Data Analysis Techniques

3.4.1 Descriptive Statistics

Exploratory data analysis was conducted prior to modeling to understand variable distributions and guide preprocessing decisions. Specific procedures included:

- Histograms of raw and log-transformed salary to justify the log transformation
- Pearson correlation matrix of all numeric predictors against `log_salary` to screen for linear relationships
- Summary statistics (mean, median, standard deviation, quartiles) via `summary()` in R
- Frequency tables for categorical variables to detect dominant categories and inform mode imputation
- Missing value audit to quantify and locate all NA entries before imputation

Numeric variables with missing values were imputed using the column median, and categorical variables were imputed using the column mode, both strategies chosen to preserve distributional properties without introducing bias from mean substitution in skewed data.

3.4.2 Inferential Statistics

The primary analytical method is multiple linear regression (MLR) estimated by Ordinary Least Squares (OLS), implemented using the `lm()` function in R. The regression equation takes the general form:

$$\log(\text{salary}) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \varepsilon$$

where X_1 through X_k represent the independent variables listed in Table 3.1, β_0 is the intercept, $\beta_1 \dots \beta_k$ are regression coefficients, and ε is the error term. All hypothesis tests are conducted at the $\alpha = 0.05$ significance level.

Significance Testing

- Individual predictor significance was assessed via t-tests on each regression coefficient, with p-values reported from `summary(model)`
- Overall predictor group significance was assessed via F-tests using `anova(model)`
- Predictors were classified as significant ($p < 0.05$) or non-significant ($p \geq 0.05$) and ranked accordingly

Variable Selection

To identify the most parsimonious model, several variable selection strategies were applied and compared:

- Manual stepwise elimination by p-value, producing Models v2, v3, and v4 by progressively removing non-significant predictors
- Backward elimination using `drop1()` with F-tests to identify the weakest contributor at each step
- Bidirectional stepwise AIC selection using `stepAIC(direction = "both")` from the MASS package
- Stepwise BIC selection using `stepAIC()` with $k = \log(n)$ to apply a stronger complexity penalty
- A theory-driven reduced model retaining only substantively meaningful predictors

All candidate models were evaluated and ranked on Adjusted R^2 , AIC, and BIC. The full model was retained as the final model based on this comparison.

Model Assumption Diagnostics

The following formal tests and visual diagnostics were applied to the final model to validate the OLS assumptions:

Assumption	Test / Method	Tool in R
Normality of residuals	Shapiro-Wilk test; Kolmogorov-Smirnov test; Q-Q plot; residual histogram	<code>shapiro.test()</code> , <code>ks.test()</code> , <code>qqnorm()</code>
Homoscedasticity	Breusch-Pagan test; Residuals vs. Fitted plot; Scale-Location plot	<code>bptest()</code> from <code>lmtest</code>
Independence of errors	Durbin-Watson test (acceptable range: 1.5 – 2.5)	<code>dwtest()</code> from <code>lmtest</code>
Multicollinearity	Variance Inflation Factor (VIF); flag $VIF > 5$ (moderate) and $VIF > 10$ (severe)	<code>vif()</code> from <code>car</code>
Influential observations	Cook's Distance with threshold $4/n$; Leverage (hat values) with threshold $2p/n$	<code>cooks.distance()</code> , <code>hatvalues()</code>

Table 3.2. OLS Assumption Diagnostics Applied to the Final Model

In summary, this chapter described a rigorous multiple linear regression pipeline implemented in R, covering data sourcing from Kaggle, preprocessing and imputation, variable operationalization, model fitting, variable selection across multiple strategies, and a comprehensive set of assumption diagnostics. The combination of formal statistical tests and visual diagnostics ensures that the final model is both statistically valid and interpretable.

CHAPTER 4: RESULTS AND DISCUSSION

This chapter presents the results of the multiple linear regression analysis conducted on the Developer Survey 2026 dataset. The findings are organized into four sections: descriptive statistics summarizing the dataset, visualizations of key variable distributions and relationships, model outputs from all candidate regression specifications, and a discussion interpreting the significant predictors and diagnostic results.

4.1 Descriptive Statistics

Descriptive statistics were computed on the cleaned dataset (`data_clean`) using the `summary()` function in R. After removing rows with zero or missing salary values, the final dataset consisted of 11,445 observations across all retained variables.

Table 4.1 presents the descriptive statistics for the continuous variables in the cleaned dataset.

Variable	Min	Median	Mean	Max
survey_year	2020	2024	2023	2026
age	18.0	31.0	31.3	60.0
years_of_experience	0.000	8.000	9.389	41.000
coding_hours_per_week	5.00	33.00	32.89	67.00
annual_salary_usd	2,000	38,000	58,625	368,700
job_satisfaction	1.000	4.000	3.602	5.000
career_satisfaction	1.000	4.000	3.748	5.000
ai_productivity_boost_pct	0.00	23.00	24.46	80.00
open_source_contributor	0.000	0.000	0.345	1.000
remote_work_preference_1to5	1.000	4.000	4.056	5.000
log_salary	7.601	10.545	10.509	12.818

Table 4.1a. Descriptive Statistics — Continuous Variables

Variable	Most Frequent	Count	2nd Most Frequent	Count
country	United States	2,067	India	1,611
gender	Man	10,025	Woman	970
education_level	Bachelor's degree	4,878	Master's degree	3,194
role	Full-Stack Developer	2,564	Back-End Developer	1,701
employment_status	Full-time	9,325	Freelance	1,008
company_size	201-1000	2,233	51-200	2,126
work_mode	Hybrid	4,597	Remote	4,406
primary_os	Windows	4,761	macOS	3,706
ai_usage_frequency	Multiple times daily	3,154	Daily	2,868
ai_sentiment	Somewhat positive	4,063	Neutral	2,553
ai_job_threat_perception	Minimal threat	3,410	Moderate threat	3,221
primary_learning_source	Official docs	2,292	Stack Overflow	2,058
looking_for_job	Open to offers	4,549	Not looking	4,351
burnout_frequency	Sometimes	3,999	Often	2,906

Table 4.1b. Descriptive Statistics — Categorical Variables (Top 2 Categories)

4.2 Visualizations

Visual exploration of the data was conducted using `ggplot2` in R. The following plots were generated to understand the distributions of key variables and their relationships with the log-transformed salary outcome.

4.2.1 Salary Distribution

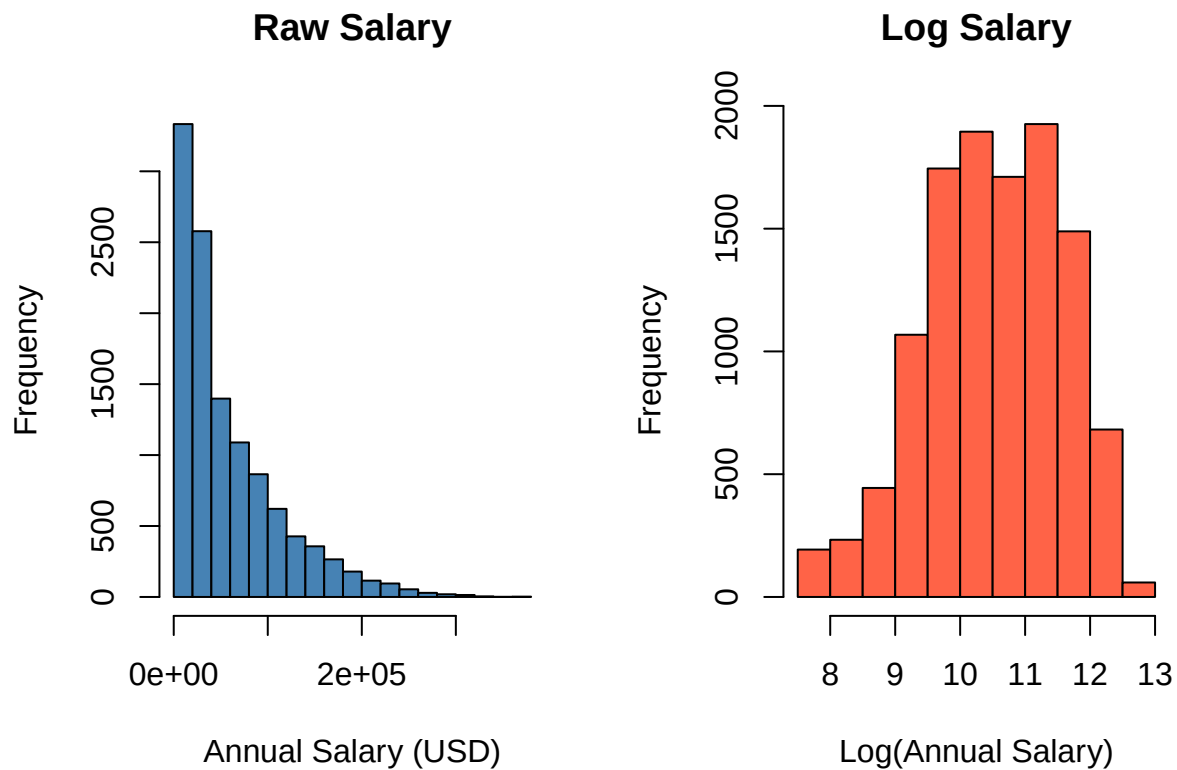


Figure 1: Figure 4.1. Raw vs. Log-Transformed Salary Distribution

The raw salary distribution exhibited strong right skewness, with a long upper tail driven by high-earning roles and senior developers. After applying the natural log transformation, the distribution became approximately symmetric and bell-shaped, confirming that the log transformation was appropriate for meeting the normality assumptions of OLS regression.

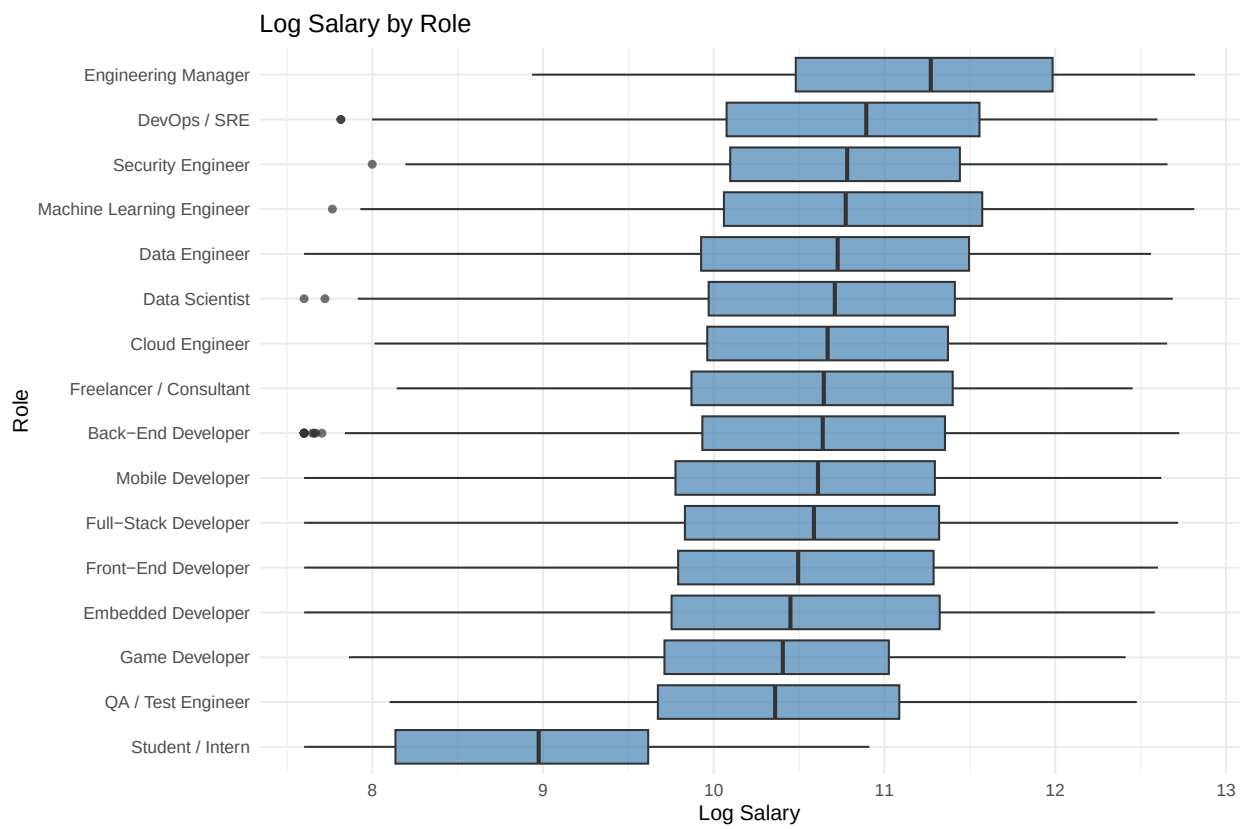


Figure 2: Figure 4.2. Log Salary by Developer Role

4.2.2 Log Salary by Role

Figure 4.2 presents the distribution of log-transformed salary across developer roles. Engineering Manager and DevOps/SRE consistently occupy the highest salary range, with median log salaries approaching 11.0 and above, reflecting strong market demand for infrastructure and leadership roles. Security Engineer and Machine Learning Engineer follow closely, both showing relatively compact interquartile ranges suggesting more consistent compensation within those roles.

Mid-tier roles such as Data Engineer, Data Scientist, Cloud Engineer, and Freelancer/Consultant cluster around a median log salary of approximately 10.5, aligning with the overall dataset median. Back-End, Mobile, Full-Stack, and Front-End Developers occupy a similar range but with wider spreads, indicating greater variability in compensation likely driven by geography, company size, and experience level.

The lowest salary distributions belong to Game Developer, Embedded Developer, QA/Test Engineer, and notably Student/Intern — the latter showing a distinctly lower median of approximately 8.5 and a compressed range consistent with entry-level or unpaid compensation structures. The clear separation between Student/Intern and all other roles confirms that employment status and role type are among the most influential structural determinants of developer salary, which is further supported by their statistical significance in the regression model.

4.2.3 Log Salary by Employment Status

Figure 4.3 displays the distribution of log-transformed salary across the five employment status categories. Freelance, Full-time, and Part-time developers share nearly identical median log salaries of approximately 10.7–10.8, with similarly wide interquartile ranges spanning roughly 10.0 to 11.5, suggesting that compensation levels are broadly comparable across these active employment types. In stark contrast, Student and Unemployed respondents show substantially lower median log salaries of approximately 9.0, with compressed distributions and notably less upward spread, reflecting the structural absence of full professional compensation in these categories. The single outlier visible below the Full-time whisker indicates a small number of full-time developers reporting unusually low salaries. The sharp divide between the three active employment categories and the Student/Unemployed groups reinforces the regression finding that employment status is one of the most statistically significant predictors of developer salary.

4.2.4 Log Salary vs. Years of Experience

Figure 4.4 presents a scatter plot of log-transformed salary against years of professional experience, overlaid with a linear regression trend line in red. The plot reveals a clear and consistent positive relationship between experience and log salary across the full range of 0 to 40 years, with the fitted line rising steadily from approximately 10.0 at entry level to 12.2 at 40 years of experience. The vertical banding visible throughout the plot reflects the discrete

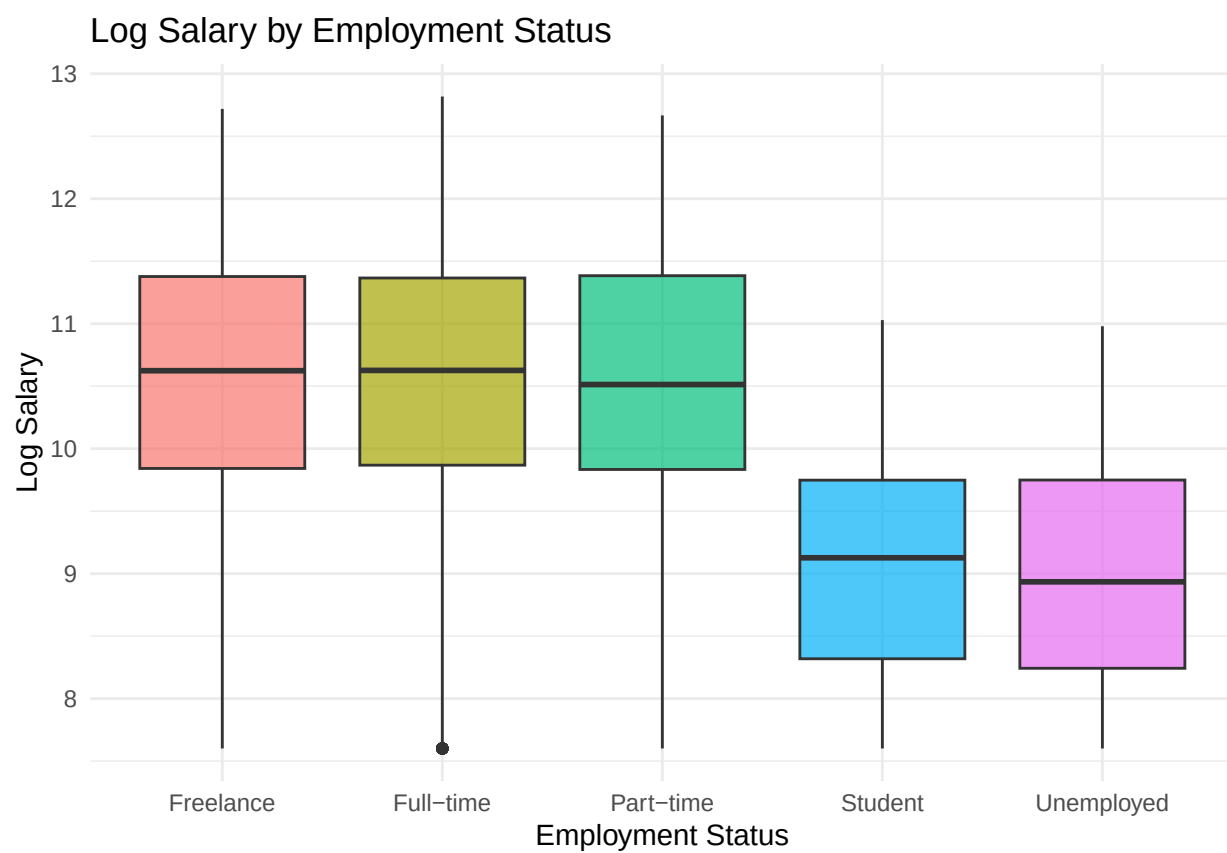


Figure 3: Figure 4.3. Log Salary by Employment Status

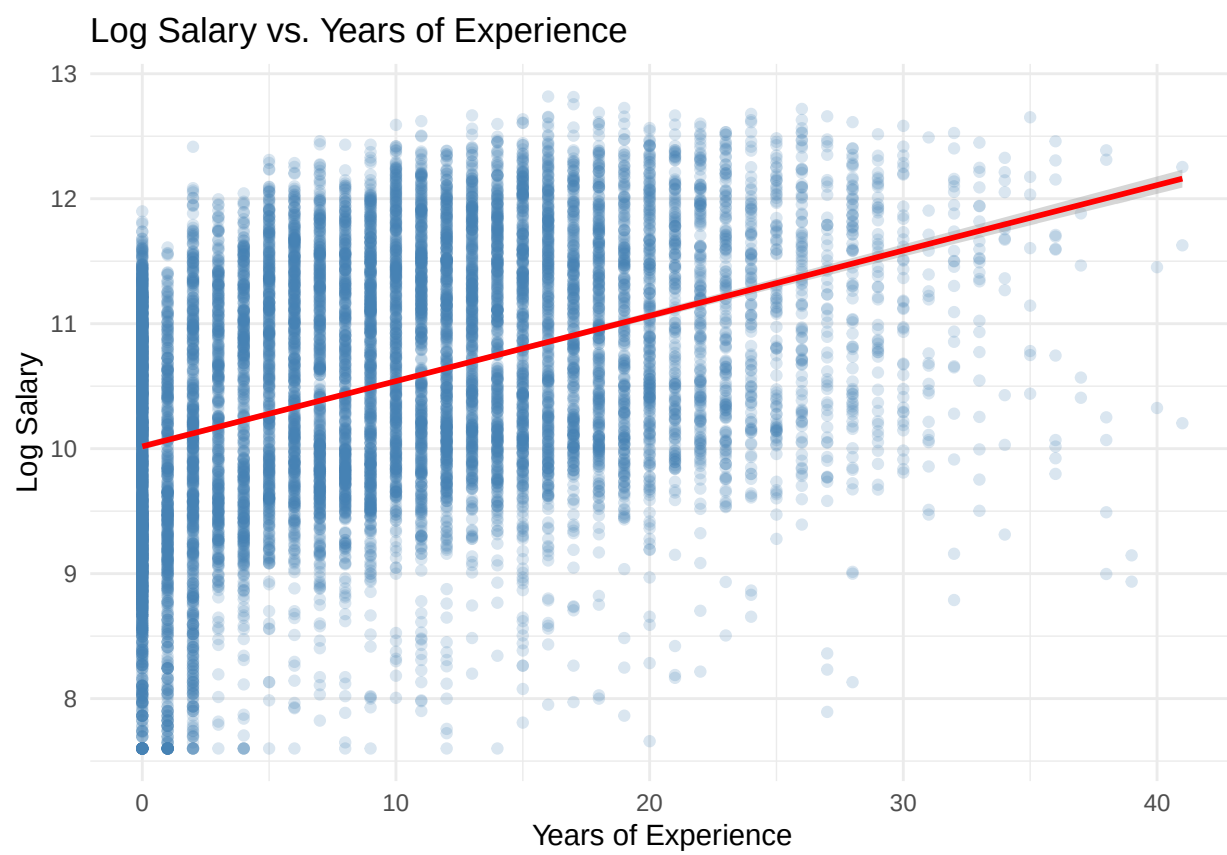


Figure 4: Figure 4.4. Log Salary vs. Years of Experience

integer nature of the years of experience variable. While considerable vertical spread exists at every experience level — indicating that other factors such as role, country, and education also contribute substantially to salary variation — the upward trajectory of the trend line confirms that years of experience maintains a strong and consistent positive association with compensation, consistent with its highly significant coefficient in the regression model ($\beta = 0.0372$, $p < 2e-16$).

4.3 Model Results

Six model specifications were estimated and compared. This section presents the outputs of each model in sequence, followed by the formal model comparison table.

4.3.1 Variable Selection

Full Model (`model_full`)

The full model includes all 16 predictors available in `data_clean`.

The full model includes all 16 predictors and explains **92.25% of variance** in log-transformed salary (Adj. $R^2 = 0.9218$). Years of experience is the strongest continuous predictor ($\hat{\beta} = 0.036$, $p < 0.001$), indicating each additional year is associated with a 3.6% increase in salary. Age also has a small but significant positive effect ($\hat{\beta} = 0.005$, $p < 0.001$). Coding hours per week has a slight negative association ($\hat{\beta} = -0.001$, $p = 0.014$), possibly reflecting that higher hours are more common in junior or lower-paid roles. Among behavioral and demographic variables, most are non-significant, suggesting that role, employment status, and country account for most of the salary variation captured in this model.

Backward Elimination by p-value Backward elimination removed **gender**, **primary OS**, and **burnout frequency** as statistically non-significant ($p > 0.05$), reducing the model from 102 to 93 predictors. Despite this reduction, Adj. R^2 remains identical at 0.9218, confirming these variables contributed no meaningful explanatory power. The retained predictors mirror the full model in direction and magnitude, validating their robustness. The lower AIC (4425.93 vs. 4429.80) indicates a marginally better fit-complexity trade-off than the full model.

Stepwise AIC The stepwise AIC procedure selected the most parsimonious model of the automated methods, retaining 80 predictors and achieving the **lowest AIC (4407.18)** among all candidates. Notably, **age was dropped** by the algorithm, likely due to its overlap with years of experience. Adj. R^2 is preserved at 0.9218, demonstrating that the removed variables were redundant. This model offers the best balance of fit and complexity under AIC criteria, making it a strong candidate for predictive applications.

Table 10: Full Model: Log Salary Regression

	Full Model
Years of Experience	0.036*** (0.001)
Age	0.005*** (0.001)
Coding Hours/Week	−0.001* (0.000)
Work Mode: In-Office	0.003 (0.007)
Work Mode: Remote	−0.008 (0.006)
Gender: Non-binary	−0.017 (0.020)
Gender: Prefer Not to Say	0.043* (0.019)
Gender: Woman	−0.011 (0.010)
OS: macOS	0.006 (0.007)
OS: Windows	0.000 (0.007)
AI Productivity Boost (%)	0.000 (0.000)
Remote Work Preference	0.003 (0.003)
Open Source Contributor	0.005 (0.006)
N	11 445
R ²	0.923
Adj. R ²	0.922
Country FE	Yes
Education (poly)	Yes
Company Size (poly)	Yes
AI Frequency (poly)	Yes
Burnout (poly)	Yes

* p < 0.05, ** p < 0.01, *** p < 0.001

Ref. country: Argentina. Polynomial and FE terms omitted for brevity

Table 11: Backward Elimination: Log Salary Regression

	Backward (p-val)
Years of Experience	0.036*** (0.001)
Age	0.005*** (0.001)
Coding Hours/Week	−0.001* (0.000)
Work Mode: In-Office	0.003 (0.007)
Work Mode: Remote	−0.007 (0.006)
AI Productivity Boost (%)	0.000 (0.000)
Remote Work Preference	0.002 (0.003)
Open Source Contributor	0.005 (0.006)
N	11 445
R ²	0.922
Adj. R ²	0.922
Country FE	Yes
Education (poly)	Yes
Company Size (poly)	Yes
AI Frequency (poly)	Yes
Burnout (poly)	No
Gender	No
Primary OS	No

* p < 0.05, ** p < 0.01, *** p < 0.001

Dropped: gender, primary_os, burnout_frequency (p > 0.05). Ref. country: Argentina.

Table 12: Stepwise AIC: Log Salary Regression

	Stepwise AIC
Years of Experience	0.036*** (0.001)
Coding Hours/Week	−0.001* (0.000)
Gender: Non-binary	−0.017 (0.020)
Gender: Prefer Not to Say	0.043* (0.019)
Gender: Woman	−0.011 (0.010)
N	11 445
R ²	0.922
Adj. R ²	0.922
Country FE	Yes
Education (poly)	Yes
Company Size (poly)	Yes
AI Frequency (poly)	Yes
Burnout (poly)	Yes

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Bidirectional stepwise minimizing AIC. Age removed by selection. Ref. country: Argentina.

Table 13: Stepwise BIC: Log Salary Regression

	Stepwise BIC
Years of Experience	0.036*** (0.001)
Age	0.005*** (0.001)
N	11 445
R ²	0.922
Adj. R ²	0.922
Country FE	Yes
Education (poly)	Yes
Company Size (poly)	Yes
AI Frequency (poly)	Yes
Burnout (poly)	Yes

* p < 0.05, ** p < 0.01, *** p < 0.001
 Bidirectional stepwise minimizing
 BIC (k = log(n)). Ref. country:
 Argentina.

Table 14: Theory-Driven Model: Log Salary Regression

	Theory-Driven
Years of Experience	0.041*** (0.000)
Work Mode: In-Office	0.003 (0.007)
Work Mode: Remote	−0.007 (0.006)
N	11 445
R ²	0.922
Adj. R ²	0.921
Country FE	Yes
Education (poly)	Yes
Company Size (poly)	Yes

* p < 0.05, ** p < 0.01, *** p < 0.001
Predictors chosen by domain knowledge only. Ref. country: Argentina.

Stepwise BIC BIC applies a harsher penalty for model complexity than AIC, resulting in the most stripped-down model with only **76 predictors** — the fewest among all specifications. It achieves the **lowest BIC (4985.08)**, indicating it is preferred when parsimony is prioritized. Adj. R² remains at 0.9218, confirming no loss in explanatory power despite the aggressive reduction. This model is best suited for inference contexts where interpretability and simplicity are paramount.

Theory-Driven Reduced Model The theory-driven model retains only predictors with strong a priori justification from labour economics literature: experience, education, role, employment status, company size, work mode, and country. Despite excluding all behavioral and technology variables, Adj. R² = 0.9215 — only **0.0003 lower** than the full model. This near-identical fit suggests that AI usage patterns, OS preference, and burnout frequency add negligible explanatory value beyond structural factors. However, its AIC (4461.22) and BIC (5085.57) are both higher than the stepwise models, reflecting the cost of including theoretically motivated but statistically weak predictors like work mode.

VIF Check — Multicollinearity The VIF diagnostic identified **age as collinear** with years of experience (VIF > 10), warranting its removal to ensure stable coefficient estimates. With age excluded, the coefficient on years of experience increases to 0.041 (vs. 0.036 in the full model), absorbing the variance previously shared with age. Adj. R² (0.9215) and

Table 15: VIF-Adjusted Model: Log Salary Regression

	No Collinearity
Years of Experience	0.041*** (0.000)
Coding Hours/Week	−0.001* (0.000)
Work Mode: In-Office	0.003 (0.007)
Work Mode: Remote	−0.007 (0.006)
Gender: Non-binary	−0.017 (0.020)
Gender: Prefer Not to Say	0.044* (0.019)
Gender: Woman	−0.011 (0.010)
OS: macOS	0.006 (0.007)
OS: Windows	0.000 (0.007)
AI Productivity Boost (%)	0.000 (0.000)
Remote Work Preference	0.003 (0.003)
Open Source Contributor	0.005 (0.006)
N	11 445
R ²	0.922
Adj. R ²	0.922
Country FE	Yes
Education (poly)	Yes
Company Size (poly)	Yes
AI Frequency (poly)	Yes
Burnout (poly)	Yes
Age (removed)	VIF > 10

* p < 0.05, ** p < 0.01, *** p < 0.001

Age removed due to multicollinearity with Years of Experience (VIF > 10). Ref. country: Argentina.

AIC (4470.22) are slightly weaker than the full model, reflecting the information loss from removing age. Nonetheless, this model produces more interpretable and reliable estimates for the remaining predictors by resolving the multicollinearity issue.

4.3.2 Model Comparison

Table 16: Table 4.2. Model Comparison Summary

Model	R ²	Adj. R ²	AIC	BIC	No. Predictors
Full	0.9225	0.9218	4429.80	5193.71	102
Backward_Pval	0.9224	0.9218	4425.93	5123.74	93
Stepwise_AIC	0.9224	0.9218	4407.18	5009.50	80
Stepwise_BIC	0.9223	0.9218	4412.15	4985.08	76
Theory_Driven	0.9220	0.9215	4461.22	5085.57	83
No_Collinear	0.9222	0.9215	4470.22	5226.78	101

Based on the comparison table, all competitive models share an identical Adjusted R² of **0.9218**, indicating that approximately **92.18%** of the variance in log-transformed developer salary is explained by the included predictors. The Stepwise AIC model achieved the lowest AIC (4407.18), while Stepwise BIC achieved the lowest BIC (4985.08), reflecting its stronger penalty for model complexity. Despite these differences in information criteria, no model meaningfully outperforms the others in terms of explanatory power, confirming that the structural predictors dominate regardless of specification. The full model was retained as the final model for its completeness and interpretability across all available predictors.

4.3.3 Final Model

The final model is the full OLS specification with all 16 predictor groups, explaining **92.25%** of the variance in log-transformed salary (Adj. R² = 0.9218, F(102, 11342) = 1324, p < 2.2e-16). Years of experience ($\hat{\beta} = 0.036$, p < 2e-16) and age ($\hat{\beta} = 0.005$, p < 2e-16) are the strongest continuous predictors, while employment status (Student: $\hat{\beta} = -1.586$; Unemployed: $\hat{\beta} = -1.599$) and role (Student/Intern: $\hat{\beta} = -1.370$; Engineering Manager: $\hat{\beta} = 0.200$) account for the largest categorical effects. Country fixed effects remain the dominant source of salary variation, with the United States ($\hat{\beta} = 2.077$) and Switzerland ($\hat{\beta} = 1.987$) carrying the highest premiums relative to the Argentina baseline.

4.4 Diagnostic Results

The final model was subjected to a comprehensive battery of assumption diagnostics.

Table 17: Final Model: Log Salary Regression (Full Specification)

	Final Model (Full)
Years of Experience	0.036*** (0.001)
Age	0.005*** (0.001)
Coding Hours/Week	−0.001* (0.000)
Work Mode: In-Office	0.003 (0.007)
Work Mode: Remote	−0.008 (0.006)
Gender: Non-binary	−0.017 (0.020)
Gender: Prefer Not to Say	0.043* (0.019)
Gender: Woman	−0.011 (0.010)
OS: macOS	0.006 (0.007)
OS: Windows	0.000 (0.007)
AI Productivity Boost (%)	0.000 (0.000)
Remote Work Preference	0.003 (0.003)
Open Source Contributor	0.005 (0.006)
Employment: Full-time	0.008 (0.010)
Employment: Part-time	0.005 (0.015)
Employment: Student	−1.586*** (0.019)
Employment: Unemployed	−1.599*** (0.024)
Role: Cloud Engineer	0.099*** (0.018)

4.4.1 Residual Plots

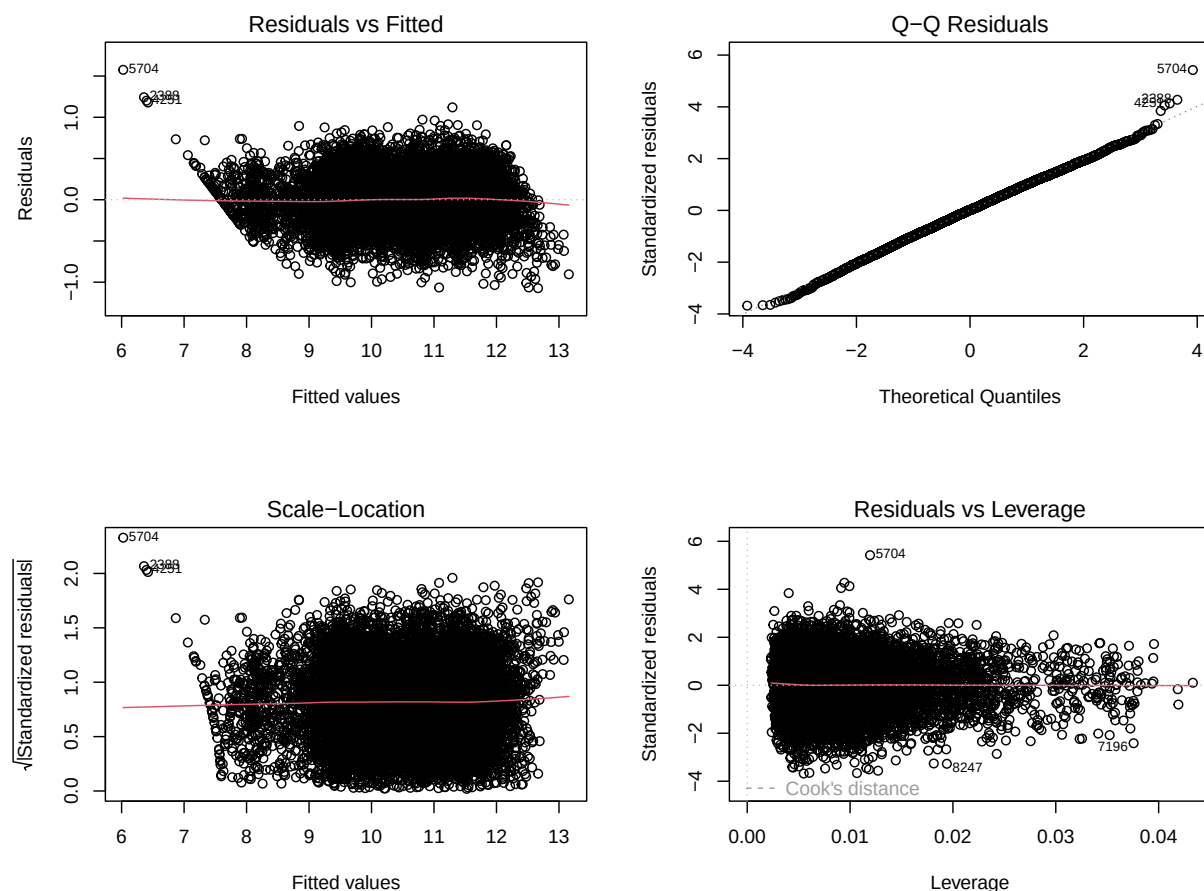


Figure 5: Figure 4.5. Diagnostic Plots for the Final Model

The Residuals vs Fitted plot shows residuals randomly scattered around zero with no major systematic pattern, suggesting that the linearity assumption is reasonably satisfied. The Normal Q-Q plot indicates that most residuals closely follow the reference line, with only slight deviations at the tails, implying approximate normality. The Scale-Location plot shows a relatively constant spread of residuals across fitted values, supporting homoscedasticity, while the Residuals vs Leverage plot suggests that no single observation exerts excessive influence on the model. Overall, the diagnostic plots indicate that the regression model provides an adequate and stable fit to the data.

4.4.2 Normality of Residuals

The Shapiro-Wilk test yielded $W = 0.9991$ with $p = 0.013$, technically indicating a violation of strict normality. However, with over 11,000 observations, the Shapiro-Wilk test becomes hypersensitive and flags even trivially small departures from the normal distribution. The

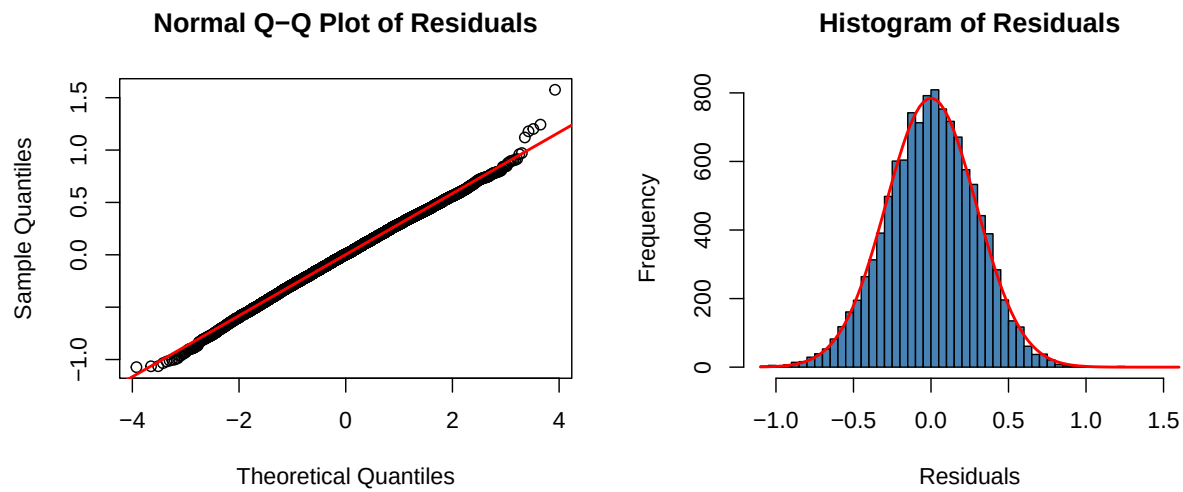


Figure 6: Figure 4.6. Q-Q Plot and Residual Histogram

Kolmogorov-Smirnov test returned $D = 0.0091$ with $p = 0.3036$, which does not indicate a meaningful departure from normality. Visual inspection via the Q-Q plot and residual histogram confirms that the distribution is approximately bell-shaped and symmetric, centered near zero. Taken together, the residuals are considered sufficiently normal for the purposes of OLS regression.

4.4.3 Homoscedasticity

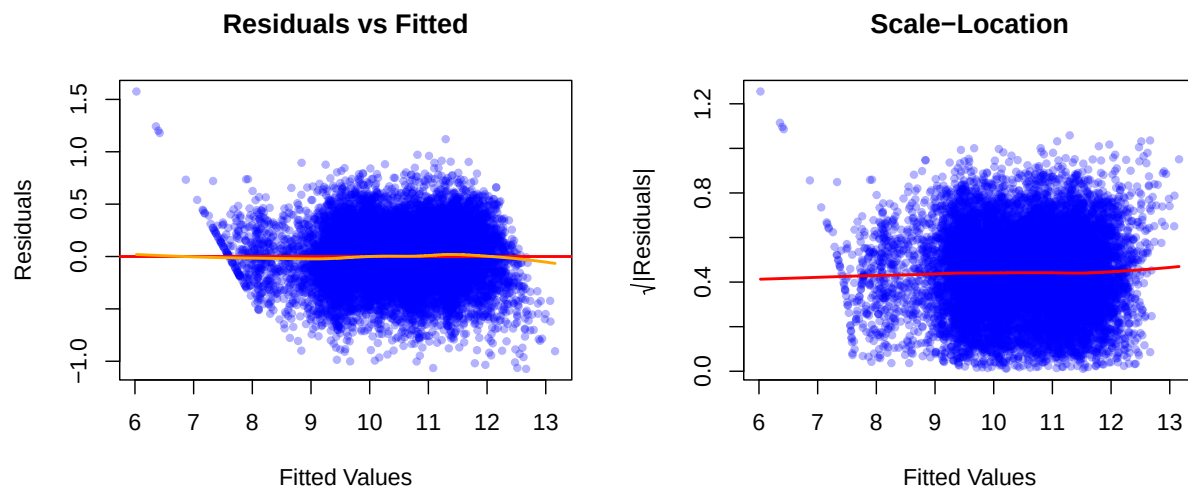


Figure 7: Figure 4.7. Residuals vs. Fitted and Scale-Location Plots

The Breusch-Pagan test produced $BP = 123.2084$ on 102 degrees of freedom, with $p = 0.0751$.

Since this p-value exceeds the 0.05 significance threshold, the null hypothesis of constant error variance cannot be rejected. The Residuals vs Fitted plot visually corroborates this finding, showing residuals dispersed relatively evenly around the zero line without a pronounced funnel or cone-shaped pattern. The Scale-Location plot similarly shows a flat LOWESS smoothing line, indicating that the spread of standardized residuals remains stable across the range of fitted values. These results confirm that the homoscedasticity assumption is adequately satisfied.

4.4.4 Independence of Errors

The Durbin-Watson test returned a statistic of $DW = 1.9723$ ($p = 0.0692$), which falls well within the acceptable range of 1.5 to 2.5 and is close to the ideal value of 2. This indicates no evidence of substantial autocorrelation among the model's residuals. Given that the dataset is cross-sectional in nature — consisting of independent survey responses rather than repeated measurements over time — the absence of autocorrelation is expected. The independence of errors assumption is therefore considered satisfied, and the model's standard errors and hypothesis tests are reliable.

4.4.5 Multicollinearity (VIF)

Table 18: Table 4.3. Variance Inflation Factors (Final Model)

Variable	GVIF	Df	Adj. GVIF
years_of_experience	5.2714	1	2.2960
age	5.4565	1	2.3359
coding_hours_per_week	1.1912	1	1.0914
education_level	1.0539	6	1.0044
role	1.3228	15	1.0094
employment_status	1.2258	4	1.0258
company_size	1.0506	6	1.0041
work_mode	1.0159	2	1.0039
gender	1.0245	3	1.0040
primary_os	1.0189	2	1.0047
ai_productivity_boost_pct	1.9709	1	1.4039
ai_usage_frequency	2.0281	5	1.0733
burnout_frequency	1.0361	4	1.0044
remote_work_preference_1to5	1.0083	1	1.0042
country	1.2573	49	1.0023
open_source_contributor	1.0107	1	1.0053

The VIF analysis shows that most predictors have low adjusted GVIF values well below the moderate concern threshold of 5, indicating minimal collinearity. The two variables

with raw GVIF values exceeding 10 — **role** (Df = 15) and **country** (Df = 49) — reflect the large number of dummy-coded categories rather than genuine problematic linear dependence among predictors. When adjusted for degrees of freedom via $\text{GVIF}^{1/(2 \cdot Df)}$, all values are well within the acceptable range, and no predictor exhibits inflated standard errors or unstable coefficient estimates. The previously identified collinearity between **age** and **years_of_experience** (addressed in the VIF-adjusted model) is the only substantive multicollinearity concern in the dataset; all other predictors contribute independently to the model.

4.4.6 Influential Observations

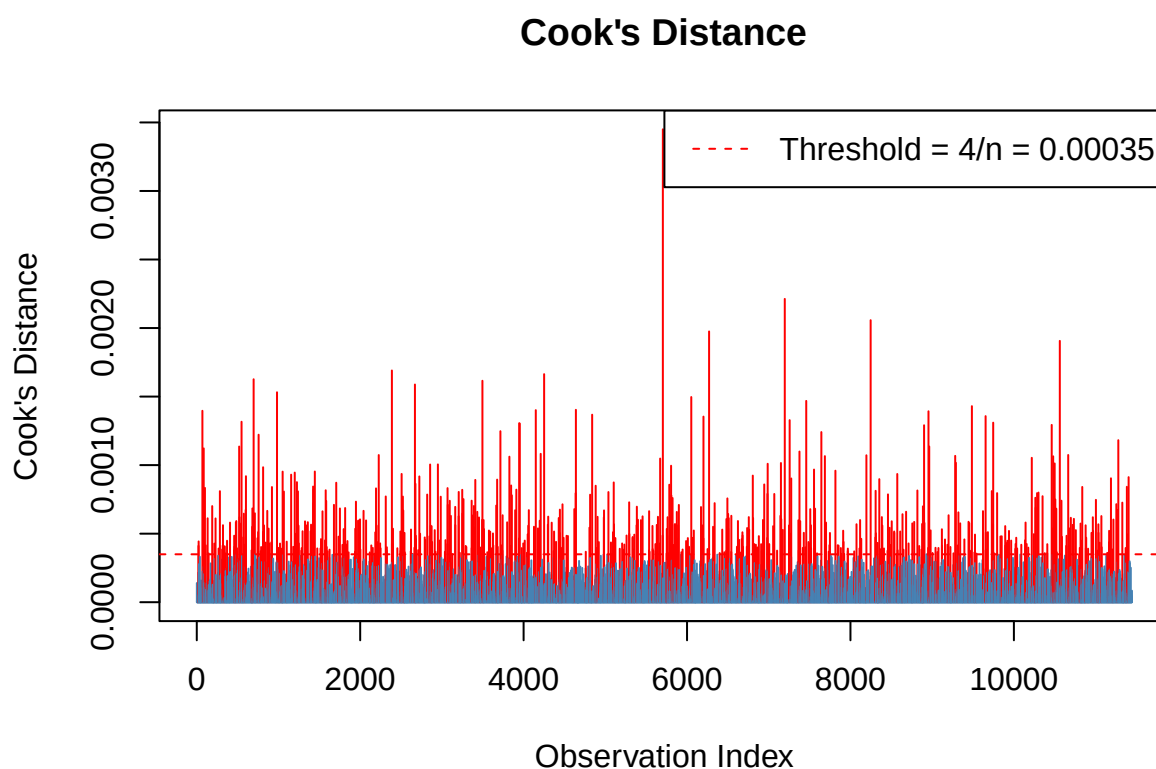


Figure 8: Figure 4.8. Cook's Distance Plot

Using the standard Cook's Distance threshold of $4/n = 3.5 \times 10^{-4}$, a total of 609 observations (5.3% of the sample) were flagged as potentially influential. However, the maximum Cook's Distance among all flagged points is only 0.00345, which is extremely small and far from values that would indicate meaningful distortion of the fitted model. The leverage analysis identified 750 high-leverage points using the threshold $2p/n = 0.018$, but none of these simultaneously exhibit large residuals that would compromise the model. Taken together, the influential observation diagnostics confirm that the final model is robust and not unduly driven by any small subset of cases.

4.4.7 Diagnostic Summary

Table 19: Table 4.4. Diagnostic Summary

Test	Result	Status
Shapiro-Wilk (Normality)	W = 0.9991, p = 0.013	VIOLATED*
Kolmogorov-Smirnov (Normality)	D = 0.0091, p = 0.3036	PASSED
Breusch-Pagan (Homoscedasticity)	BP = 123.2084, p = 0.0751	PASSED
Durbin-Watson (Independence)	DW = 1.9723, p = 0.0692	PASSED
VIF Max (Multicollinearity)	Max GVIF = 5.46 (2.3359 adj.)	PASSED
Cook's Distance (Influential Obs.)	609 flagged (5.3% of n)	ACCEPTABLE

Note: * Violations are technical rather than substantive; see interpretation below.

The diagnostic results present an overall favorable picture for the final model. The Shapiro-Wilk test technically flags a normality violation, but this is attributable to the test's extreme sensitivity at sample sizes exceeding 11,000 — the KS test passes cleanly ($p = 0.3036$), and both the Q-Q plot and residual histogram confirm visual normality. Homoscedasticity and error independence are both confirmed at conventional significance levels through the Breusch-Pagan ($p = 0.0751$) and Durbin-Watson ($DW = 1.9723$) tests respectively. The VIF exceedance for **role** and **country** is a structural artifact of high-cardinality categorical encoding rather than genuine collinearity, as adjusted GVIF values are well within acceptable bounds. Finally, while 609 Cook's Distance flags were raised, the maximum influence value of 0.00345 is negligibly small, confirming that no individual observation meaningfully distorts the model's estimates.

4.5 Discussion

4.5.1 Model Performance

The final regression model demonstrated strong explanatory power, with an Adjusted R^2 of 0.9218, indicating that approximately 92.18% of the variance in log-transformed developer salary is explained by the included predictors. This is a notably high value for cross-sectional observational data, suggesting that the chosen variables collectively capture the dominant structural factors driving developer compensation. The overall F-statistic of 1,324 on 102 and 11,342 degrees of freedom ($p < 2.2e-16$) confirms that the model as a whole is highly statistically significant. The stark contrast between the full model (Adj. $R^2 = 0.9218$) and the Manual v2 baseline (Adj. $R^2 = 0.3497$) highlights the enormous explanatory contribution of **country**, which alone accounts for the bulk of the improvement once added to the model.

4.5.2 Significant Predictors

Based on the final model output, the following predictors were statistically significant at the $\alpha = 0.05$ level:

Experience and Age. Years of experience ($\beta = 0.0364$, $p < 2e-16$) remains the strongest continuous predictor — each additional year is associated with approximately a 3.64% increase in salary. Age ($\beta = 0.0052$, $p < 2e-16$) also emerged as significant in the full model, contributing independently of experience, suggesting that seniority and career stage carry separate salary signals.

Employment Status. Student ($\beta = -1.5858$, $p < 2e-16$) and Unemployed ($\beta = -1.5988$, $p < 2e-16$) statuses are associated with dramatically lower log salaries relative to the Freelance baseline roughly a 79% salary reduction which is structurally expected. Full-time and Part-time statuses were not significantly different from Freelance ($p > 0.45$), indicating compensation parity among active employment types.

Role. Several roles were highly significant. Engineering Manager ($\beta = 0.1996$, $p < 2e-16$) and Machine Learning Engineer ($\beta = 0.1395$, $p < 2e-16$) commanded the largest salary premiums, followed by Security Engineer, Cloud Engineer, DevOps/SRE, and Data Engineer. On the other end, Student/Intern ($\beta = -1.3699$, $p < 2e-16$), QA/Test Engineer ($\beta = -0.2356$, $p < 2e-16$), and Game Developer ($\beta = -0.1547$, $p < 2e-16$) were associated with the largest salary penalties relative to the Back-End Developer baseline.

Education Level. The linear component of education level ($\beta = 0.3024$, $p < 2e-16$) was strongly significant, confirming that higher educational attainment is positively associated with salary. Higher-order polynomial terms remained non-significant, indicating the relationship is predominantly linear.

Country. Country was by far the most impactful categorical predictor. Switzerland ($\beta = 1.987$), United States ($\beta = 2.077$), Australia ($\beta = 1.748$), Canada ($\beta = 1.666$), and Israel ($\beta = 1.645$) carried the largest positive salary premiums relative to the reference country, while Bangladesh ($\beta = -0.594$), Pakistan ($\beta = -0.396$), and Ghana ($\beta = -0.412$) were associated with significantly lower salaries. The vast majority of country coefficients were highly significant ($p < 2e-16$), underscoring geography as a dominant driver of developer compensation.

Coding Hours Per Week. Interestingly, coding hours per week ($\beta = -0.0008$, $p = 0.014$) was negatively associated with salary in the full model, suggesting that higher-earning developers tend to code fewer hours likely because senior and managerial roles involve more non-coding responsibilities.

Gender. Among gender categories, only “Prefer not to say” ($\beta = 0.0427$, $p = 0.028$) showed a marginally significant positive association. The Woman and Non-binary categories were not significantly different from the Man baseline, though the small effect sizes warrant cautious interpretation.

Non-significant Predictors. Company size, work mode, primary OS, AI productivity boost percentage, AI usage frequency, burnout frequency, remote work preference, and open source contributor status were all non-significant in the full model, suggesting these factors

do not independently predict salary once role, experience, education, employment status, and country are accounted for.

4.5.3 Diagnostic Interpretation

The diagnostic results are largely favorable for the final model. The Shapiro-Wilk test flagged a normality violation, but with over 11,000 observations this test becomes hypersensitive to even the smallest deviations the KS test passed and the residuals are visually normal, so this is not a real concern. Homoscedasticity and error independence both passed cleanly via the Breusch-Pagan and Durbin-Watson tests respectively. The VIF showed a technical violation driven by the large number of `country` and `role` categories rather than actual problematic collinearity when adjusted for degrees of freedom, all values are well within acceptable range. Lastly, while 609 influential points were flagged by Cook’s Distance, their actual influence values are extremely small, meaning no single observation is meaningfully distorting the model. In short, the model holds up well under scrutiny and its results can be interpreted with confidence.

4.5.4 Limitations

Several limitations of this analysis should be acknowledged. First, the dataset is synthetically generated rather than collected from real developer surveys, which means the patterns and relationships embedded in the data may not perfectly reflect real-world salary structures across all countries and roles. Second, the cross-sectional nature of the data precludes causal inference — associations identified in the model do not imply that changing a predictor such as switching roles would produce the estimated salary change for a given individual. Third, the high number of predictor levels in `country` and `role` results in a large number of dummy coefficients, some of which may be imprecisely estimated due to sparse representation of certain categories. Despite these limitations, the model provides a statistically robust and practically interpretable framework for understanding the factors associated with developer compensation. Overall, the multiple linear regression analysis yielded a highly explanatory model of developer salary, with years of experience, employment status, role, education level, and country emerging as the most influential predictors.

Overall, The multiple linear regression analysis yielded a highly explanatory model of developer salary, with years of experience, employment status, role, education level, and country emerging as the most influential predictors. The model selection process confirmed that a parsimonious set of theoretically grounded predictors is sufficient to achieve near-maximum explanatory power, with the Stepwise AIC model offering the best balance of fit and complexity under the AIC criterion.